

Macroeconomics

ECON 101

Money, Banks, and the Bank of Canada
(Chapter 10)

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Chapter Outline

10.1 What Is Money, and Why Do We Need It?

10.2 How Is Money Measured in Canada Today?

10.3 How Do Banks Create Money?

10.5 The Quantity Theory of Money

Money

Money is one of the most important inventions of mankind.

- Economists consider **money** to be any asset that people are generally willing to accept in exchange for goods and services, or for payment of debts.
- **Asset:** Anything of value owned by a person or a firm.

We will begin by considering what role money serves, and what can be used as money.

- Then we will consider modern forms of money and the roles of banks and the government in creating and managing money.
- Finally, we will create a model that relates prices to the amount of money.

What Is Money, and Why Do We Need It?

10.1 Learning Objective

Define money and discuss the four functions of money.

Suppose you were living before the invention of money.

- If you wanted to trade, you would have to **barter**, trading goods and services directly for other goods and services.
- Trades would require a **double coincidence of wants**.

Eventually, societies started using **commodity money**—goods used as money that also have value independent of their use as money—like animal skins or precious metals.

- The existence of money makes trading much easier and allows specialization, an important step for developing an economy.

The Functions of Money

Money fulfills four primary functions:

Medium of exchange

Money is acceptable to a wide variety of parties as a form of payment for goods and services.

Unit of account

Money allows a way of measuring value in a standard manner.

Store of value

Money allows people to defer consumption till a later date by storing value. Other assets can do this too, but money does it particularly well because it is *liquid*, easily exchanged for goods.

Standard of deferred payment

Money facilitates exchanges *across time* when we anticipate that its value in the future will be predictable.

What Can Serve as Money?

In order to serve as an acceptable medium of exchange (and hence a potential “money”), an asset should have the following characteristics:

1. The asset must be *acceptable* to most people.
2. It should be of *standardized quality* so any two units are alike.
3. It should be *durable* so that value is not lost by wearing out.
4. It should be *valuable* relative to its weight, so that it can easily be transported even in large quantities.
5. It should be *divisible* to be used in purchases of both low-priced and high-priced goods.

Commodity Money

Commodity money has a value independent of its use as money.

Some important historical and modern commodity moneys:

- Precious metals, such as gold or silver
- Beaver pelts in early settlers of members of an Indigenous group in Canada
- Cigarettes in prisons and prisoner-of-war camps

Fiat Money (1 of 2)

In modern economies, paper money is generally issued by a *central bank* run by the government.

The **Bank of Canada** is the central bank of Canada. However, money issued by the Bank of Canada is no longer exchangeable for gold; nor is *any* current world currency. Instead, the Bank of Canada issues currency known as **fiat money**.

Fiat money refers to any money, such as paper currency, that is authorized by a central bank or governmental body, and that does not have to be exchanged by the central bank for gold or some other commodity money.

Fiat Money (2 of 2)

Fiat money: Money, such as paper currency, that is authorized by a central bank or governmental body and that does not have to be exchanged by the central bank for gold or another commodity money.

- It is more flexible for central banks to create money.

However it also creates a potential problem: fiat money is only acceptable as long as *households and firms have confidence that if they accept paper dollars in exchange for goods and services, the dollars will not lose much value during the time they hold them.*

- If people stop “believing” in the fiat money, it will cease to be useful.

How Is Money Measured in Canada Today?

10.2 Learning Objective

Discuss the definitions of the money supply used in Canada today.

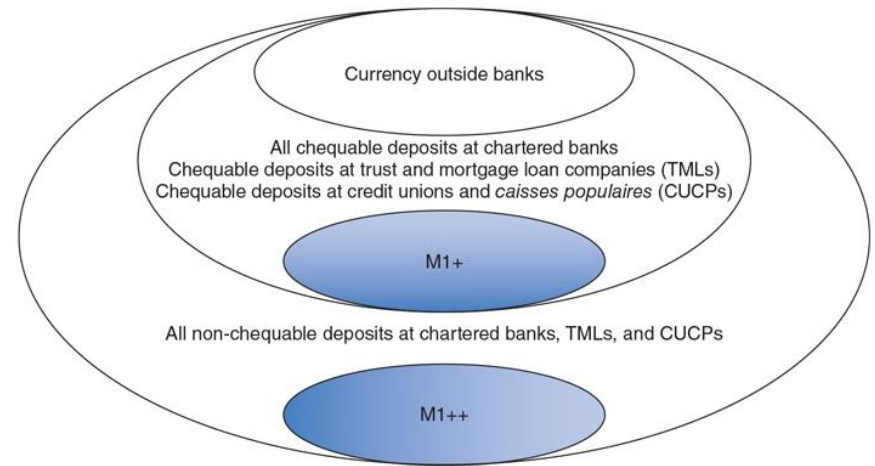
The Bank of Canada has conducted several studies to determine the appropriate measure of money in the economy.

Figure 10.1 Components of the Money Supply (1 of 3)

M1+ is the narrowest definition of money it includes:

1. Currency in circulation
2. Chequable deposits at banks, trust and mortgage companies, credit unions and caisses populaires.

M1++ includes M1+ and non-chequable deposits



(a) Components of M1+ and M1++
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The Bank of Canada uses six different definitions of the money supply: M1+, M1++, M2, M2+, M2++, and M3. M1+ is the narrowest definition of the money supply and M2++ is the broadest.

Figure 10.1 Components of the Money Supply (2 of 3)

M2 includes

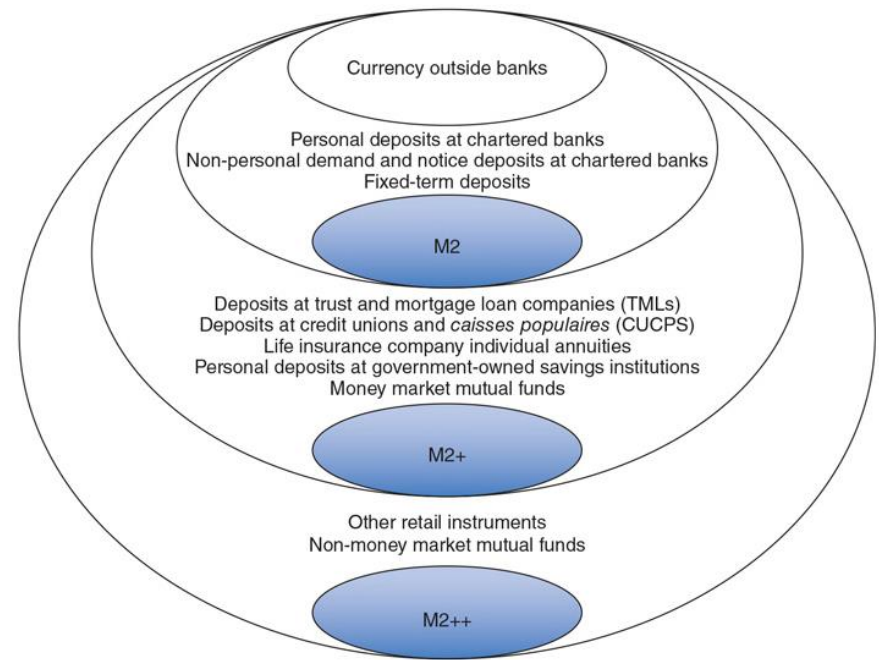
- Currency
- personal deposits at banks
- non-personal demand and notice deposits at banks
- fixed term deposits.

M2+ includes M2 and

- Deposits at TMLs and CUCPs
- life insurance companies individual annuities
- Government owned savings institutions
- Money market mutual funds.

M2++ includes (broadest definition) M2+

- Canada Savings Bonds
- Non-money market mutual funds.

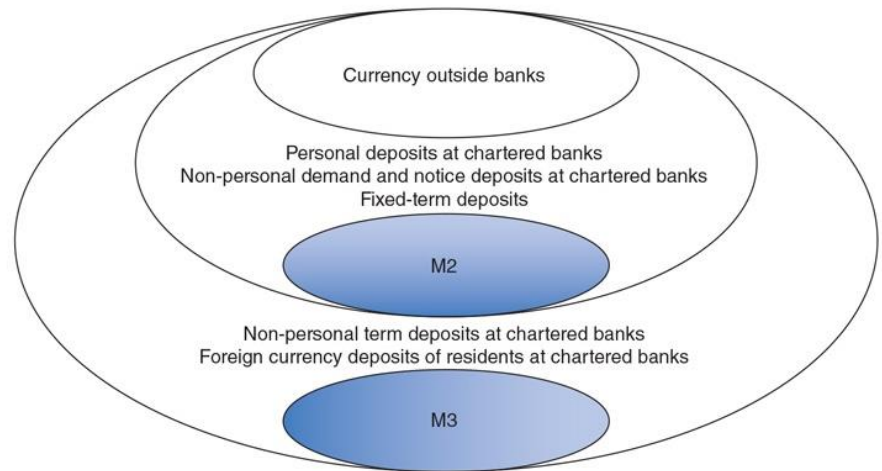


(b) Components of M2, M2+, and M2++
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Figure 10.1 Components of the Money Supply (3 of 3)

M3 includes

- M2
- Non-personal term deposits at banks
- Foreign currencies deposits of Canadians at banks



(c) Components of M3
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Which Definition?

When we want to talk about the money supply, which definition should we use?

Any one of the three might be valid, but we are mostly interested in money's role as the medium of exchange, so this suggests using M1+.

In our discussion of money, we will therefore:

1. Treat both currency and chequing and non-chequing account balances as “money”, but nothing else.
2. Realize that banks play an important role in the money supply, since they control what happens to money when it is in an account.

How Do Banks Create Money?

10.3 Learning Objective

Explain how banks create money.

Banks play a critical role in the money supply.

- Recall that there is more money held in checking accounts than there is actual currency in the economy.
- So money is being *created* by banks.

Further, banks are profit-making private firms: among the largest corporations in the country.

- Their activities are designed to allow themselves to make a profit.

In order to understand the role that banks play, we will first try to understand how banks operate as a business.

Figure 10.2 Balance Sheet for Royal Bank, September 30, 2021 (1 of 2)

Asset (in billions)		Liabilities and Shareholders' Equity (in billions)	
Reserves	\$199	Deposits	\$1082
Loans	950	Borrowing	24
Securities	282	Other liabilities	531
Other assets	302		
		Total liabilities	\$1637
		Shareholders' equity	96
Total assets	\$1733	Total liabilities and shareholders' equity	\$1733

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SOURCE: <http://www.osfi-bsif.gc.ca/Eng/Pages/default.aspx> (accessed on December 9, 2021).

On a balance sheet, a firm's assets are listed on the left, and its liabilities (and stockholders' equity, or **net worth**) are listed on the right. The left and right sides must add to the same amount.

- Banks use money deposited with them to make loans and buy securities (investments).
- Their largest liabilities are their deposit accounts: money they owe to their depositors.

Figure 10.2 Balance Sheet for Royal Bank, September 30, 2021 (2 of 2)

Asset (in billions)		Liabilities and Shareholders' Equity (in billions)	
Reserves	\$199	Deposits	\$1082
Loans	950	Borrowing	24
Securities	282	Other liabilities	531
Other assets	302	Total liabilities	\$1637
		Shareholders' equity	96
Total assets	\$1733	Total liabilities and shareholders' equity	\$1733

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Reserves are deposits that a bank keeps as cash in its vault or on deposit with the Bank of Canada.

- *Notice that the bank does not keep enough deposits on hand to cover all of its deposits. This is a big part of how the bank makes a profit: lending out or investing money deposited with it.*

Required and Excess Reserves

The bank must keep *some* cash available for its depositors; it does this through a combination of *vault cash* and deposits with the Bank of Canada.

These funds are **reserves**: deposits that a bank keeps on hand to meet customers' withdrawals needs.

Banks in Canada are **NOT** required to hold a specific level of reserves.

Banks desire to keep a fraction of their deposits (about 5 percent) on hand as reserves.

This 5% is known as the **desired reserve ratio (r_d)**: the fraction of deposits banks want to keep as reserves.

Banks might increase their desired reserve ratio or reduce it depending on the state of the economy or state of financial markets.

Using T-Accounts to Show Money Creation

A T-account is a stripped-down version of a balance sheet, showing only how a transaction *changes* a bank's balance sheet.

When you deposit \$1,000 in currency at Bank of Montreal (BMO), its reserves increase by \$1,000 and so do its deposits:

Assets		Liabilities	
Reserves	+\$1000	Deposits	+\$1000

Your deposit of \$1000 into your chequing account increases Bank of Montreal's assets and liabilities by the same amount.

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The currency component of the money supply decreases by the \$1,000, since that \$1,000 is no longer in circulation; but the checking deposits component increases by \$1,000. So there is no net change in the money supply—yet.

T-Accounts

But BMO needs to make a profit; so it keeps 10% (for example) of the deposit as reserves, and lends out the rest, creating a \$900 checking account deposit.

Assets		Liabilities	
Reserves	+\$1000	Deposits	+\$1000
Loans	+\$900	Deposits	+\$900

1. By loaning out \$900 in excess reserves . . .

2. . . . Bank of Montreal has increased the money supply by \$900.

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The \$900 initially appears in a BMO checking account but will soon be spent; and BMO will transfer \$900 in currency to the bank at which the \$900 check is deposited.

Bank of Montreal			
Assets		Liabilities	
Reserves	+\$100	Deposits	+\$1000
Loans	+\$900		

1. When the \$900 cheque that was deposited in a Royal Bank account arrives to be cleared, the increase in Bank of Montreal's reserves (shown in the previous T-account) falls by \$900 to \$100 . . .

2. . . . and the increase in Bank of Montreal's deposits falls by \$900 to \$1000.

Royal Bank			
Assets		Liabilities	
Reserves	+\$900	Deposits	+\$900

After the cheque drawn on the account at Bank of Montreal clears, Royal Bank's reserves and deposits both increase by \$900.

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When Will It End?

Each “round”, the additional checking account deposits get smaller and smaller.

- Every round, 10% of the deposits are kept as reserves. This allows us to tell by how much the checking deposits will eventually increase: the \$1,000 in currency will become the 10% required reserves for all of the checking deposits, so a total of \$10,000 in checking deposits can be created.

Bank	Increase in Chequing Account Deposit
Bank of Montreal	\$1000
Royal Bank	+ 900 (= $0.9 \times \$1000$)
Third Bank	+ 810 (= $0.9 \times \$900$)
Fourth Bank	+ 729 (= $0.9 \times \$810$)
•	+ •
•	+ •
•	+ •
Total change in chequing account deposits	= \$10 000

Simple Deposit Multiplier

An alternative way to find out how much money the original \$1,000 in currency will create is to add up all of the checking account deposits.

$$\$1000 + [0.9 \times \$1000] + [(0.9 \times 0.9) \times \$1000] + [(0.9 \times 0.9 \times 0.9) \times \$1000] + \dots$$

$$\$1000 + [0.9 \times \$1000] + [0.9^2 \times \$1000] + [0.9^3 \times \$1000] + \dots$$

$$\$1000 \times (1 + 0.9 + 0.9^2 + 0.9^3 + \dots)$$

- The expression in the parentheses can be rewritten as:

$$\frac{1}{1-0.9} = \frac{1}{0.10} = 10$$

- So the total increase in deposits is $\$1,000(10) = \$10,000$.
- The “10” here is the **simple deposit multiplier**: the ratio of the amount of deposits created by banks to the amount of new reserves.

General Form for the Simple Deposit Multiplier

In general, we can write the simple deposit multiplier as:

$$\text{Simple deposit multiplier} = \frac{1}{r_d}$$

So with a 10% desired required reserve ratio (r_d), the simple deposit multiplier is 10.

- With a 20% desired required reserve ratio, the simple deposit multiplier is 5.

$$\text{Change in chequing account deposits} = \text{Change in bank reserves} \times \frac{1}{r_d}$$

For example, \$100,000 in new deposit, with a 10% desired reserve ratio, results in:

$$\begin{aligned}\text{Change in checking account deposits} &= \$100,000 \times \frac{1}{0.10} \\ &= \$100,000 \times 10 = \$1,000,000\end{aligned}$$

The Simple Deposit Multiplier vs Real-World Deposit Multiplier

With a 10% desired reserve ratio, the simple deposit multiplier tells us that a currency deposit will be multiplied 10 times.

- But in reality, we do not observe this.

Why this difference?

- Banks may not lend out as much as we predict, either because they want to keep *excess reserves*, or they cannot find credit-worthy borrowers.
- Consumers keep some currency out of the bank; that currency cannot be used as required reserves.

Note: during the financial crisis of 2007–2009, research suggests that the real-world multiplier fell to close to 1 in the US.

Conclusions About Banks and the Money Supply

In general, we can assume that the real-world deposit multiplier is greater than 1. So we conclude that:

1. When banks gain reserves, they make new loans, and the money supply expands.
2. When banks lose reserves, they reduce their loans, and the money supply contracts.

This is enough to establish the important relationship between banks and the money supply.

The Quantity Theory of Money

10.5 Learning Objective

Explain the quantity theory of money and use it to explain how high rates of inflation occur.

Beginning in the 16th century, Spain sent gold and silver from Mexico and Peru back to Spain.

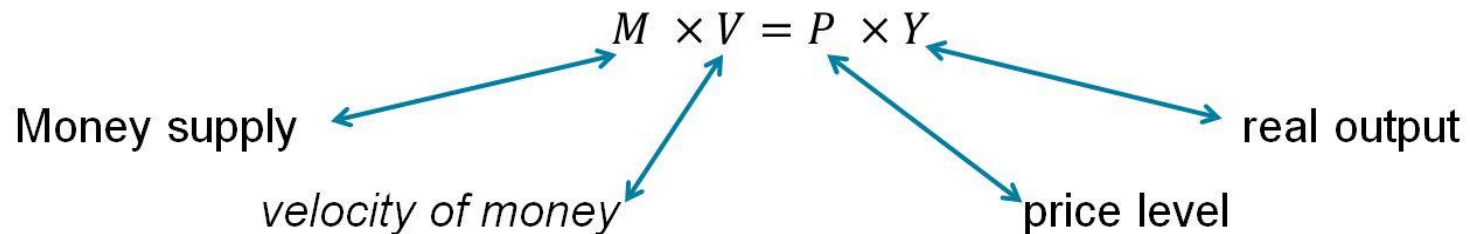
- These metals were minted into coins, increasing the money supply.

Prices in Europe rose steadily during those years.

- This helped observers make the connection between inflation and the flow of gold and silver into Europe from the Americas.

The Quantity Equation

In the early 20th century, Irving Fisher formalized the relationship between money and prices by using *the quantity equation*:



Velocity of money: the average number of times per year each dollar in the money supply is used to purchase goods and services included in GDP.

Rewriting this equation by dividing through by M, we obtain:

$$V = \frac{P \times Y}{M}$$

Calculating the Velocity of Money

Measuring:

- The money supply (M) with M1+,
- The price level (P) with the GDP deflator, and
- The level of real output (Y) with real GDP,

We obtain the following value for velocity (V) in 2021:

$$V = \frac{(\$2.5 \text{ trillion})}{(\$1.5 \text{ trillion})} = 17$$

We can always calculate V. But will we always get the same answer?

The **quantity theory of money** is a theory about the connection between money and prices that assumes that the velocity of money is constant.

The Quantity Theory Explanation of Inflation

When variables are multiplied together in an equation, we can form the same equation with their *growth rates added together*.

So the quantity equation:

$$M \times V = P \times Y$$

generates:

$$\begin{aligned} &\text{Growth rate of the money supply} + \text{Growth rate of velocity} \\ &= \text{Growth rate of the price level (or the inflation rate)} \\ &+ \text{Growth rate of real output} \end{aligned}$$

We can rewrite the equation to help understand what determines inflation by assuming that velocity of money is constant, we obtain:

$$\text{Inflation rate} = \text{Growth rate of the money supply} - \text{Growth rate of real output}$$

The Inflation Rate According to the Quantity Theory

Inflation rate

= Growth rate of the money supply – Growth rate of real output

This equation provides the following predictions:

1. If the money supply grows faster than real GDP, there will be inflation.
2. If the money supply grows slower than real GDP, there will be *deflation* (a decline in the price level).
3. If the money supply grows at the same rate as real GDP, there will be neither inflation nor deflation: the price level will be stable.

Is velocity truly constant from year to year? The answer is *no*.

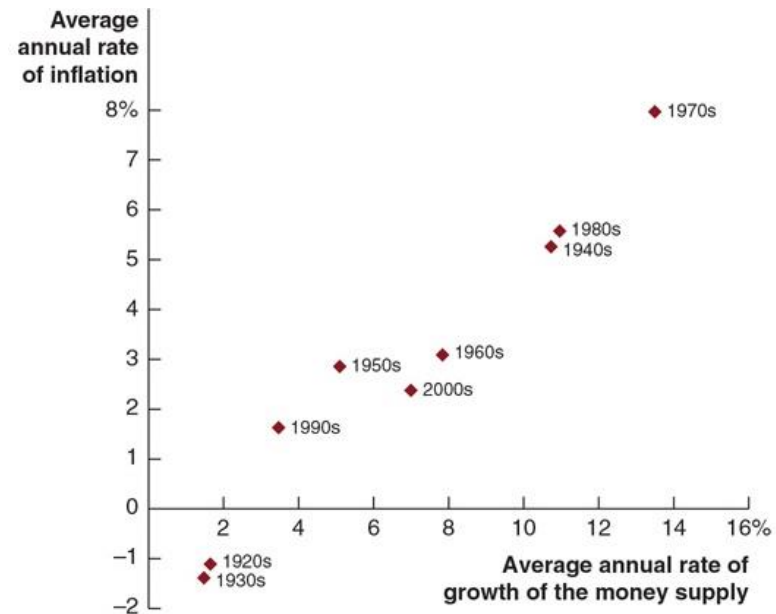
- But the quantity theory of money can still provide insight:
- In the long run, inflation results from the money supply growing at a faster rate than real GDP.

Figure 10.7 The Relationship between Money Growth and Inflation over Time and around the World (1 of 2)

Real GDP growth has been relatively consistent over time.

So based on the quantity theory of money (QTM), there should be a predictable, positive relationship between the annual rates of inflation and growth rates of the money supply.

There *is* a positive relationship, more stronger in the long run.



(a) Inflation and money supply growth in Canada, 1920s–2010s

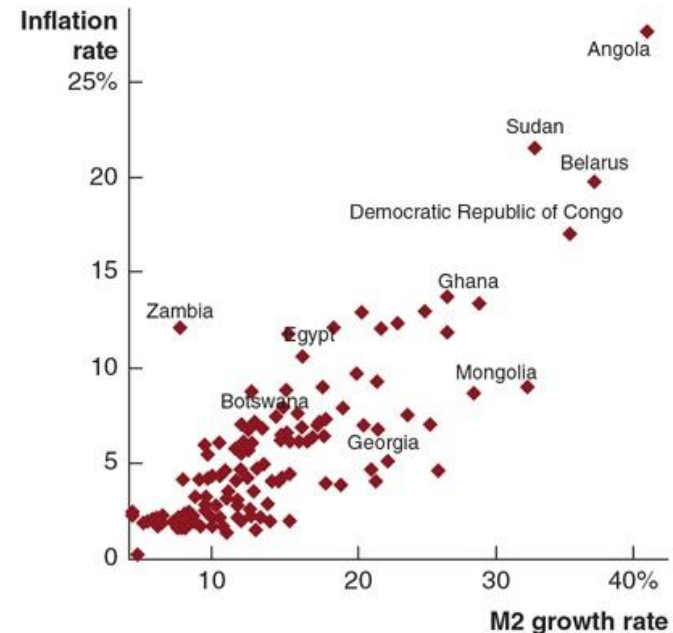
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SOURCES: Panel (a): Data for 1870 to 1960, M. C. Urquhart, “Canadian Economic Growth 1870–1985,” Institute for Economic Research, Queens University, Kingston, Ontario, Discussion Paper No. 734; and since 1961, Statistics Canada, CANSIM II series V3860248 and V41552796

Figure 10.7 The Relationship between Money Growth and Inflation over Time and around the World (2 of 2)

We see a similar story when we compare average rates of inflation and growth rates of the money supply across different countries.

Although the relationship is not entirely predictable, countries with higher growth in the money supply do have higher rates of inflation.



(b) Inflation and money supply growth in 129 countries, 2002–2018

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SOURCES: Panel (b): International Monetary Fund, Money and Financial Statistics.

High Rates of Inflation

Very high rates of inflation—in excess of 50 percent per month—are known as *hyperinflation*.

- **Hyperinflation** results when central banks increase the money supply at a rate far in excess of the growth rate of real GDP.
- This might happen when governments want to spend much more than they raise through taxes, so they force their central bank to “buy” government bonds.

Recently, hyperinflation has occurred in Zimbabwe; the inflation rate became high enough, people stop using paper currency.

Hyperinflation tends to be associated with slow growth, if not severe recession.